

SQL Queries - Math

Sometimes you want to apply *Math concepts to your SQL queries*, to better get information from your data.

- This part of the lab is going to be quite light & simple.

These functions all sit *alongside the "SELECT"*

Firstly, **MAX** (and with it obviously, **MIN**)

```
SELECT MAX(yearformed) -- Largest Number
FROM Artist
WHERE yearended IS NOT NULL

SELECT MIN(yearformed) -- Smallest Number
FROM Artist
```

	(No column name)
1	2019
	(No column name)
1	1996

Notice how it says *No column name*, we can save these **PINK** functions.

- Using **AS** afterwards.

```
SELECT MAX(yearformed) AS NewestBand
FROM Artist

SELECT MIN(yearformed) AS OldestBand
FROM Artist
```

	NewestBand
1	2019
	OldestBand
1	1996

We can also check for AVERAGES,

```
SELECT AVG(yearformed) AS averageformation  
FROM Artist
```

	averageformation
1	2008

The last 3 functions were for numbers only, but what if I want to *count*?
It makes sense to be able to count the number of times a word happens?

```
SELECT COUNT(Country) AS USArtists  
FROM Artist  
WHERE Country = 'USA'|
```

	USArtists
1	3

Obviously also, SUM.

```
SELECT SUM(sales) AS totalAlbumSales  
FROM Album
```

Which outputs →

	totalAlbumSales
1	11550000

Then; What if, I want to count every **DISTINCT** country of any Artist?

```
SELECT COUNT(DISTINCT country) AS UniqueArtistCountries
FROM Artist
```

DISTINCT allows for filtering for **unique entries**

- *(This can be coupled with other functions too).*

The **last thing**, you can actually do math operations **INSIDE** the query.

- For every album, let's get the *amount of sales* for **each track** in their album.

```
SELECT albumtitle,
       sales / tracks AS salespertrack
FROM Album
```

	albumtitle	salespertrack
1	Midnight Rain	192307
2	Sunrise Over	150000
3	Starlight	213333
4	Summer Pop	90000
5	Dance Hits	56250
6	Euphoria	92857
7	Pop Revolution	91666
8	Classic Vibes	30000

*That's it for this part of the lab, next part will go over
relating different tables & keys.*